MAPS BETWEEN PULLED BACK OBJECTS ARE COMPUTED BY A COLIMIT ALONG A DESCENDING SYSTEM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The comparison map for the union of two opens satisfies the Mayer-Vietoris condition, so the induction principle applies. invoked at proof lines 7466-7475. Reduces the statement to quasi-compact opens and their unions.
- On an affine scheme a perfect or pseudo-coherent complex is represented by a bounded above complex of finite free modules, and a complex of finite tor dimension is described by a tor amplitude bound. invoked at proof lines 7476-7529. Supplies the concrete representatives with which the colimit computation is performed in the affine case.
- The statement over the ring, which computes maps between the pulled back complexes as a colimit, is contained in the algebra chapter. invoked at proof lines 7500-7529. Finishes the affine case and hence the proof.

Source reference. Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 7466-7470 to organise the proof; the Mayer-Vietoris verification and the affine case are inside the counted span.

Source reference. Original lemma 08BW: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 7472-7476 for the union step of the induction; the reduction to the affine case that follows is inside the counted span.

Source reference. Original lemma 08EB: sources/stacks-perfect/pinned-original-source.tex, source lines 2636–2636. Used at proof lines 7480-7490 to represent the perfect object in the first case; the rest of the affine computation is inside the counted span.

Source reference. Original lemma 08E7: sources/stacks-perfect/pinned-original-source.tex, source lines 2440–2440. Used at proof lines 7492-7500 to represent the pseudo-coherent object in the second case; the finishing argument is inside the counted span.

Source reference. Original lemma 08E9: sources/stacks-perfect/pinned-original-source.tex, source lines 2508–2508. Used together with the previous label at proof

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lines 7495-7500 to handle the finite tor dimension hypothesis; the translation to algebra is inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Cited at proof lines 7485-7490 in the affine reduction; the computation that uses it is inside the counted span.

Source reference. Original lemma 0654: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 7500-7510 as part of the algebra input that finishes the affine case; the reduction to it is inside the counted span.

Lemma 0.2. *In Situation 0.3. Let E_0 and K_0 be objects of $D(\mathcal{O}_{S_0})$. Set $E_i = Lf_{i0}^*E_0$ and $K_i = Lf_{i0}^*K_0$ for $i \geq 0$ and set $E = Lf_0^*E_0$ and $K = Lf_0^*K_0$. Then the map*

$$\operatorname{colim}_{i\geq 0} \operatorname{Hom}_{D(\mathcal{O}_{S_i})}(E_i, K_i) \longrightarrow \operatorname{Hom}_{D(\mathcal{O}_S)}(E, K)$$

is an isomorphism if either

1. *E_0 is perfect and $K_0 \in D_{QCoh}(\mathcal{O}_{S_0})$, or*
2. *E_0 is pseudo-coherent and $K_0 \in D_{QCoh}(\mathcal{O}_{S_0})$ has finite tor dimension.*

Situation 0.3. Let $S = \lim_{i\in I} S_i$ be a limit of a directed system of schemes with affine transition morphisms $f_{i'i} : S_{i'} \to S_i$. We assume that S_i is quasi-compact and quasi-separated for all $i \in I$. We denote $f_i : S \to S_i$ the projection. We also fix an element $0 \in I$.

Proof. For every open $U_0 \subset S_0$ consider the condition P that the canonical map

$$\operatorname{colim}_{i\geq 0} \operatorname{Hom}_{D(\mathcal{O}_{U_i})}(E_i|_{U_i}, K_i|_{U_i}) \longrightarrow \operatorname{Hom}_{D(\mathcal{O}_U)}(E|_U, K|_U)$$

is an isomorphism, where $U = f_0^{-1}(U_0)$ and $U_i = f_{i0}^{-1}(U_0)$. We will prove P holds for all quasi-compact opens U_0 by the induction principle of Cohomology of Schemes, Lemma 08DR. Condition (2) of this lemma follows immediately from Mayer-Vietoris for hom in the derived category, see Cohomology, Lemma 08BW. Thus it suffices to prove the lemma when S_0 is affine.

Assume S_0 is affine. Say $S_0 = \operatorname{Spec}(A_0)$, $S_i = \operatorname{Spec}(A_i)$, and $S = \operatorname{Spec}(A)$. We will use Lemma 06Z0 without further mention.

In case (1) the object $E_0^\bullet$ corresponds to a finite complex of finite projective A_0-modules, see Lemma 08EB. We may represent the object K_0 by a K-flat complex $K_0^\bullet$ of A_0-modules. In this situation we are trying to prove

$$\operatorname{colim}_{i\geq 0} \operatorname{Hom}_{D(A_i)}(E_0^\bullet \otimes_{A_0} A_i, K_0^\bullet \otimes_{A_0} A_i) \longrightarrow \operatorname{Hom}_{D(A)}(E_0^\bullet \otimes_{A_0} A, K_0^\bullet \otimes_{A_0} A)$$

Because $E_0^\bullet$ is a bounded above complex of projective modules we can rewrite this as

$$\operatorname{colim}_{i\geq 0} \operatorname{Hom}_{K(A_0)}(E_0^\bullet, K_0^\bullet \otimes_{A_0} A_i) \longrightarrow \operatorname{Hom}_{K(A_0)}(E_0^\bullet, K_0^\bullet \otimes_{A_0} A)$$

Since there are only a finite number of nonzero modules E_0^n and since these are all finitely presented modules, this map is an isomorphism.

In case (2) the object E_0 corresponds to a bounded above complex $E_0^\bullet$ of finite free A_0-modules, see Lemma 08E7. We may represent K_0 by a finite complex $K_0^\bullet$ of flat A_0-modules, see Lemma 08E9 and More on Algebra, Lemma 0654. In particular $K_0^\bullet$ is K-flat and we can argue as before to arrive at the map

$$\operatorname{colim}_{i\geq 0} \operatorname{Hom}_{K(A_0)}(E_0^\bullet, K_0^\bullet \otimes_{A_0} A_i) \longrightarrow \operatorname{Hom}_{K(A_0)}(E_0^\bullet, K_0^\bullet \otimes_{A_0} A)$$

It is clear that this map is an isomorphism (only a finite number of terms are involved since $K_0^\bullet$ is bounded). □